Rakan Eyad

Robotics & Embedded Systems Engineer

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SUMMARY

Mechatronic engineering graduate with hands-on experience across robotics, embedded systems, and control theory: from an FPGA-driven assistive robot and a ROS2 TurtleBot, to a self-built closed-loop tracking system, to migrating an industry prototype's architecture onto a Raspberry Pi.

EXPERIENCE

System Engineer — ALBON Nov 2024 — Jan 2026

- Developed and integrated hardware, software, and mechanical components for a prototype control system: migrated the architecture from a microcontroller to a Raspberry Pi for real-time data processing, integrated sensor arrays, and engineered the system's electrical junction box to cut power consumption and keep the hardware reliable.

Engineering Intern — Escape The Room Sep 2021 — Nov 2021

- Designed mechanical puzzles and implemented engineering concepts in real-world, client-facing products.

Testing Technician Intern — CloudAk May 2021 — Aug 2021

- Supported server system testing and maintenance in data-centre environments.

FEATURED PROJECTS

Autonomous Target Tracking System — Personal build · hardware

- Designed and built a self-contained laser targeting system that autonomously locks onto and tracks moving targets in real time.
- Tuned sensor calibration and signal filtering for precise, low-latency actuation; extended the system toward predictive targeting from target velocity and trajectory.

Assistive Robot (FPGA) — University build · Verilog

- Designed a Bluetooth-controlled assistive robot on an FPGA in Verilog, integrating multiple sensors into closed-loop control for autonomous operation.

Autonomous TurtleBot (ROS2) — University build · ROS2

- Programmed an autonomous TurtleBot in ROS2 to identify, count, and interact with items via QR codes, combining sensor fusion with path planning.

Planetary Rover RL Navigation — Thesis · Isaac Sim

- Developed an Honours thesis on autonomous rover navigation, pairing an A* global planner with a deep-RL local policy trained in NVIDIA Isaac Sim on Mars terrain.
- Implemented a fast A* command term and terrain importer for hybrid planning; tuned SAC training configs and reward/termination logic across 1000s of parallel environments; built a custom evaluation pipeline.

SKILLS

Firmware & electronics: C / C++ · Arduino, FPGA · Verilog / SystemVerilog, Raspberry Pi · microcontrollers, Sensor integration & calibration, Power wiring & junction boxes, Assembly

Robotics & simulation: ROS2, Isaac Sim / Isaac Lab, Sensor fusion & path planning, HIL simulation, CAD (Fusion/SW), Mechanism design

Control & signals: Advanced control systems, Signal processing & filtering, State estimation, GNC fundamentals, MATLAB, Radar & antenna coursework

EDUCATION

B.E. (Honours), Mechatronic Engineering

University of Sydney · 2022 — 2026

EWAM (Engineering Weighted Average Mark) 76% — Distinction